NEXORA · AI & DATA SCIENCE

Requirements & Equipment

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The right blood unit, to the right hospital, before it's requested... and by drone when there's no time

Everything the project needs: functional and non-functional requirements, equipment and costs, technology, permits, partners, team and a submission checklist.

Team	**NEXORA**
Field	Health · Artificial Intelligence & Data Science · Drones
Competition	Jordan Students Pitch Competition — GEW Jordan
University / Supervisor	.....
Team members	.....

> **Because blood that arrives late... doesn't arrive.**

1 Functional requirements

ID	Requirement
FR-01	Show live stock for each type at each site, with the expiry date of every unit.
FR-02	Alert automatically when stock drops below the safety level or nears expiry.
FR-03	Forecast 14-day demand per type and hospital with a confidence band.
FR-04	Propose transferring units between hospitals before they expire.
FR-05	Create an emergency request (hospital, type, quantity, priority) from the app.
FR-06	Rank possible sources by availability, distance and arrival time.
FR-07	Compare drone and car arrival times, accounting for peak traffic.
FR-08	Launch a drone mission only after approval by a licensed operator.
FR-09	Track position, altitude, speed, battery and box temperature live.
FR-10	Alarm immediately if temperature leaves 2–6 °C, with a fallback action.
FR-11	Confirm receipt and store the full temperature log for every shipment.
FR-12	Select eligible donors by type, distance and last donation, and send invitations.
FR-13	Mass-casualty mode: automatically distribute simultaneous requests across the fleet.
FR-14	Impact reports: units saved, minutes saved, share of safe shipments.

2 Non-functional requirements

Security & privacy	Encryption in transit and at rest, role-based access, explicit donor consent.
Reliability	99.9% availability for the emergency service, backups, and a manual fallback on any failure.
Performance	Source suggestion in under 5 seconds, tracking updated every second.
Auditability	Tamper-proof log of every request, decision and flight.
Usability	Arabic/English interface; an emergency request in fewer than 4 taps.
Integration	APIs for existing blood bank and hospital systems.

3 Equipment and estimated cost

Item	Qty	Unit cost (est.)	Total
Medical multirotor drone, 3–5 kg payload	2	JOD 11,000	JOD 22,000
Smart cooling box with phase-change material and temperature logger	4	JOD 750	JOD 3,000
Spare batteries and fast charger	1	JOD 2,000	JOD 2,000
Landing pad and ground control station	1	JOD 3,000	JOD 3,000
Software platform and forecasting model (development)	1	JOD 15,000	JOD 15,000
Donor app (development)	1	JOD 8,000	JOD 8,000
Cybersecurity and compliance	1	JOD 5,000	JOD 5,000
Total			**JOD 58,000**

4 Drone and cooling-box specification

Type	Multirotor drone for the pilot, then fixed-wing VTOL for longer distances
Payload	2–4 kg: 4–6 blood units with the cooled box
Range & speed	15–40 km for urban pilots, at 60–90 km/h
Cooling box	Thermal insulation and phase-change material holding 2–6 °C, with a temperature logger and live alarm
Connectivity	4G/5G cellular with backup link, streaming position, temperature and battery every second
Safety	Emergency parachute, automatic return on link loss, and geofencing
Delivery	Landing on a dedicated pad on the hospital roof or a small-parachute drop in a safe zone
Operations	A licensed pilot monitors from the operations center, on pre-approved routes

5 Technology stack

Front end	Web dashboard + mobile donor app (React / Flutter)
Back end	Python (FastAPI) + PostgreSQL + real-time streaming over WebSocket
AI	Time-series models (Prophet / LightGBM) for forecasting, and mathematical optimization for source selection and fleet dispatch
Drone	Integration via MAVLink with the ground station, plus restricted-zone maps
Cooling sensors	IoT temperature loggers reporting every second over cellular
Hosting	Cloud inside Jordan or the health authority's servers, per data-protection requirements

6 Required data

- Blood bank inventories
- Request history
- Donor database (with consent)
- Weather and airspace data
- Temperature sensor readings from transport boxes

7 Permits and approvals

Authority	Required
Civil Aviation Regulatory Commission	Drone operating permit, approval of routes and licensed pilot, starting with a pilot route.
Relevant security authorities	Approvals for flights over residential areas and near sensitive sites.
Ministry of Health & National Blood Bank	Partnership agreement and compliance with transfusion and cold-chain standards.
Data protection	Privacy policy, donor consent and compliance with the Personal Data Protection Law.
Insurance	Civil liability insurance on the fleet and shipments.
Company registration	Register a startup and protect the name and trademark.

8 Required partners

- Ministry of Health & National Blood Bank
- Private and public hospitals, Royal Medical Services
- Medical drone operators and manufacturers
- Civil Aviation Regulatory Commission & security authorities
- Jordan Red Crescent
- Telecom and insurance companies, CSR sponsors

9 Team and roles

Project Lead / AI Lead	Technical architecture, forecasting models and coordination
Data Scientist	Stock and demand data, and model evaluation
Systems Engineer / Drones & IoT	Drone, cooling box and sensor integration
App Developer	Dashboard, donor app and live demo
Business & Pitch Lead	Health and regulatory partnerships, business model and pitch

10 Submission checklist

- ☐ Full proposal (PDF)
- ☐ Business Model Canvas with all nine blocks
- ☐ Financial, technical and legal feasibility study
- ☐ Requirements and equipment document
- ☐ Pitch deck PDF + editable PPTX
- ☐ 5-minute pitch script and expected judge questions
- ☐ Interactive simulator/app for the live demo
- ☐ Fill in university, supervisor and team names on the cover
- ☐ Check the official competition rules and deadlines with the university

11 Demo-day preparation

- ☐ Charged laptop with the simulator open in advance (works offline)
- ☐ PDF of the deck on a backup USB stick
- ☐ Rehearse the live scenario 3 times before going in
- ☐ Assign roles: who speaks and who runs the simulator
- ☐ Printed copies of the BMC and summary for the judges
- ☐ A timer to stay within 5 minutes

Financial figures are preliminary planning assumptions set by the team, not market research or supplier quotes, and must be validated before any decision. “Not available in Jordan” means no equivalent was found in preliminary research. The simulator uses synthetic data and schematic locations.